



**Aror University of Art, Architecture, Design and Heritage**  
**Faculty of Artificial Intelligence and Multimedia and Gaming**

**Work Sheet: 04**

Q: 01 Convert each degree measure to radian measure and radian measure to degree.

1.  $120^\circ$    2.  $-20^\circ$    3.  $-60^\circ$    4.  $144^\circ$    5.  $\frac{3\pi}{2}$    6.  $-\frac{7\pi}{6}$    7.  $-\frac{7\pi}{12}$

Q: 02 Find the length of the arc on a circle of radius 15 inches intercepted by a central angle 120.

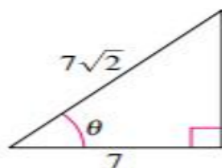
Q: 03 Find the radian measure of the central angle of a circle of radius 80 km that intercepts an arc of length 150 km.

Q: 04 Evaluate the six trigonometric functions at the real number.

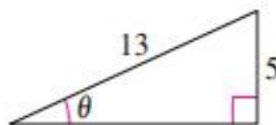
1.  $t = \frac{3\pi}{2}$    2.  $t = -\frac{7\pi}{6}$    3.  $t = \frac{\pi}{3}$

Q: 05 Find the exact values of the six trigonometric functions of the angle  $\theta$ .

1.



2.



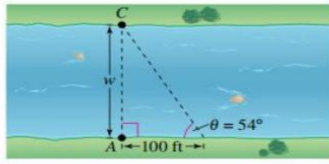
Q: 06 Find the exact values of the other five trigonometric functions of  $\theta$ .

1.  $\sin\theta = \frac{3}{4}$    2.  $\cos\theta = 3$    3.  $\sec\theta = \frac{2}{3}$

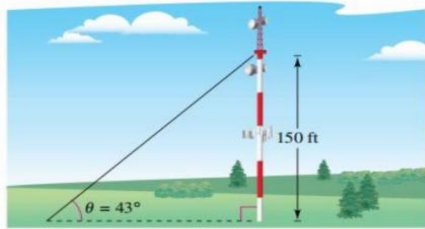
Q: 07 Use trigonometric identities to transform the left side of the equation into the right side.

1.  $\tan\theta \cos\theta = \sin\theta$    2.  $(1 + \sin\theta)(1 - \sin\theta) = \cos^2\theta$   
 3.  $(\sec\theta + \tan\theta)(\sec\theta - \tan\theta) = 1$    4.  $\sin^2\theta - \cos^2\theta = 2\sin^2\theta - 1$   
 5.  $\frac{\tan\theta + \cot\theta}{\tan\theta} = \operatorname{cosec}^2\theta$    6.  $\frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta} = \operatorname{cosec}\theta \sec\theta$    7.  $\tan^2 x \sec^2 x - \tan^2 x = \tan^4 x$

Q: 08 A biologist wants to know the width  $w$  of a river to properly set instruments for an experiment. From point A, the biologist walks downstream 100 feet and sights to point C (see figure). From this sighting, it is determined that  $\theta = 54^\circ$ . How wide is the river?



Q: 09 A guy wire runs from the ground to a cell tower. The wire is attached to the cell tower 150 feet above the ground. The angle formed between the wire and the ground is  $43^\circ$  (see figure).



(a) How long is the guy wire?

(b) How far from the base of the tower is the guy wire anchored to the ground?

Q: 10 The point is on the terminal side of an angle in standard position. Find the exact values of the six trigonometric functions of the angle.

1.  $(5, 12)$
2.  $(8, 15)$
3.  $(-5, -2)$

Q: 11 Determine the quadrant in which  $\theta$  lies.

1.  $\sin \theta > 0, \cos \theta > 0$
2.  $\sin \theta < 0, \cos \theta < 0$
3.  $\csc \theta > 0, \tan \theta < 0$
4.  $\sec \theta > 0, \cot \theta < 0$

Q: 12 Find the exact values of the remaining trigonometric functions of  $\theta$  satisfying the given conditions.

1.  $\tan \theta = \frac{15}{8}, \sin \theta > 0$
2.  $\cos \theta = \frac{8}{17}, \tan \theta < 0$
2.  $\sin \theta = 0.6, \theta$  lies in Quadrant II.
3.  $\cos \theta = -0.8, \theta$  lies in Quadrant III.

Q: 13 Find the reference angle  $\theta'$ . Sketch  $\theta$  in standard position and label  $\theta'$ .

1.  $\theta = 160^\circ$     2.  $\theta = 309^\circ$     3.  $\theta = -125^\circ$     4.  $\theta = -215^\circ$     5.  $\frac{2\pi}{3}$     6.  $\frac{7\pi}{6}$

Q: 14 Sketch the graphs of  $f$  and  $g$  in the same coordinate plane.

1.  $f(x) = \sin x$ ,  $g(x) = \sin \frac{x}{3}$     2.  $f(x) = \cos x$ ,  $g(x) = 2 - \cos x$

3.  $f(x) = \cos x$ ,  $g(x) = \cos^{-1} x$     4.  $f(x) = \sin x$ ,  $g(x) = \sin^{-1} x$

Q15: A ladder that is 16 feet long leans against the side of a house. The angle of elevation of the ladder is  $80^\circ$ . Find the height from the top of the ladder to the ground.

Q16: Use the fundamental identities to simplify the expression.

1.  $\tan^2 x - \tan^2 x \sin^2 x$     2.  $\sec^2 x \sin^2 x - \sin^2 x$     3.  $\frac{\tan x \cot x}{\sec x}$     4.  $\frac{\sin x}{1 + \cos x} + \frac{\sin x}{1 - \cos x}$

Q17: Find exact value of each expression.

1.  $\cos(\frac{\pi}{4} + \frac{\pi}{3})$     2.  $\sin(135^\circ - 30^\circ)$     3.  $\tan(60^\circ + 45^\circ)$

4.  $\cos \frac{\pi}{3} \cos \frac{\pi}{4} - \sin \frac{\pi}{3} \sin \frac{\pi}{4}$     5.  $\frac{\tan 30^\circ + \tan 45^\circ}{1 - \tan 30^\circ \tan 45^\circ}$

6.  $\sin 195^\circ + \sin 105^\circ$     7.  $\cos 80^\circ - \cos 35^\circ$

Q18: Use a double-angle formula to solve the equations.

1.  $\sin 2x - \sin x = 0$     2.  $\sin 2x \sin x = \cos x$     3.  $\tan 2x - 2 \cos x = 0$